

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

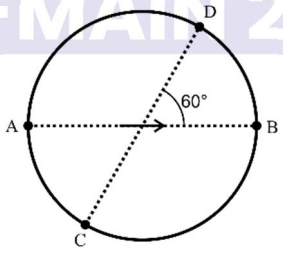
Sec: SR STAR - A & APEX, ALL IITS **DT: 24 /03 /2026**
Time: 3 Hrs **GTM - 03** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A dipole of dipole moment p is kept at the centre of a circle of radius r as shown in the figure. The radius of the circle is very large in comparison to the distance between the two charges of the dipole. A & B are two points on the axis and C & D are two points on the line at 60° angle with axis of dipole as shown. If V_A, V_B, V_C and V_D are potentials at A, B, C and D respectively then, which of the following is correct?



(A) $V_A - V_B = \frac{kp}{r^2}$, $V_C - V_D = \frac{kp}{r^2}$

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(B) $V_A - V_B = \frac{kp}{r^2}$, $V_C - V_D = \frac{2kp}{r^2}$

(C) $V_A - V_B = \frac{2kp}{r^2}$, $V_C - V_D = \frac{kp}{2r^2}$

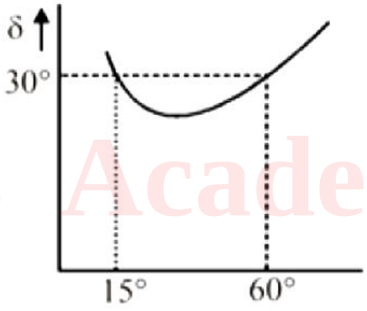
(D) $V_B - V_A = \frac{2kp}{r^2}$, $V_D - V_C = \frac{kp}{r^2}$

2. A cylindrical wire of copper having area of cross section 2.5 mm^2 is carrying 1 Ampere current. What will be the change in mobility if the current is made 4 Ampere and cross section of the copper wire is changed to 1.25 mm^2? (Temperature remains constant)

(A) Mobility will become 4 times (B) Mobility will become 8 times

(C) Mobility will become half (D) Mobility will remain same

3. Figure shows graph of deviation δ versus angle of incidence for a light ray striking a prism. Angle of prism is___



A) 30° B) 45° C) 60° D) 75°

4. If the electric potential on the axis of an electric dipole at a distance 'r' from it is V, then the potential at a point on its equatorial line at the same distance away from it will be

A) 2 V B) V/2 C) 0 D) –V

5. An organ pipe closed at one end has a harmonic with a frequency of 500 Hz. The next higher harmonic in a organ pipe closed at one end but of double length has frequency 350 Hz. Which of the following option is correct? (Speed of sound in air is 340 m/s)

(A) length of first organ pipe is 170 cm and it was vibrating in 5th overtone

(B) length of first organ pipe is 85 cm and it was vibrating in 5th overtone

(C) length of first organ pipe is 85 cm and it was vibrating in 5th harmonic

(D) length of first organ pipe is 170 cm and it was vibrating in 5th harmonic

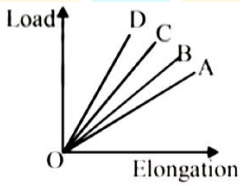
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6. A wave is propagating along x-axis. The displacement of particles of the medium in z-direction. At $t=0$ is given by : $z=\exp\left[-(x+2)^2\right]$, where '$x$' is in meters. At $t=1s$, the same wave disturbance is given by $z=\exp\left[-(2-x)^2\right]$. Then, the wave propagation velocity is

A) 4 m/s in + x direction B) 4 m/s in –x direction

C)2 m/s in + x direction D)2 m/s in-x direction

7. The load versus strain graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line .



A) OB B) OA C) OD D) OC

8. Two uniform solid spheres A and B of same material, painted completely black and placed in free space separately where no radiation is incident on spheres. Their radii are R and 2R respectively and the dominating wavelengths (wavelength corresponding to which spectral emissive power is maximum) in their spectrum are observed to be in the ratio 1 : 2. Which of the following is not correct ?

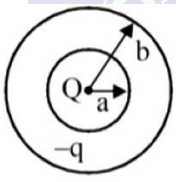
(A) Ratio of their temperatures is 2 : 1

(B) Ratio of their emissive powers is 4 : 1

(C) Ratio of their rates of heat loss is 4 : 1

(D) Ratio of their rates of cooling is 32 : 1

9. Charge – q is distributed on the surfaces as



A) –Q on the inner surface, -q on outer surface

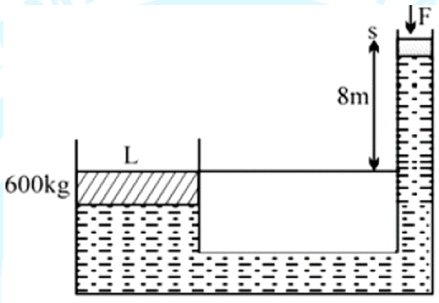
B) –Q on the inner surface, -q+Q on the outer surface

C) +Q on the inner surface, -q –Q on the outer surface

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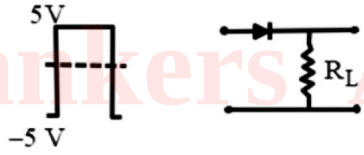
D) The charge –q is spread uniformly between the inner and outer surface.

10. For the system shown in the figure, the cylinder on the left at L has a mass of 600kg and a cross sectional area of 800 cm^2. The piston on the right, at S, has cross sectional area 25 cm^2 and negligible weight. If the apparatus is filled with oil. $\left(\rho = 0.75\ gm/cm^3\right)$ Find the force F required to hold the system in equilibrium.

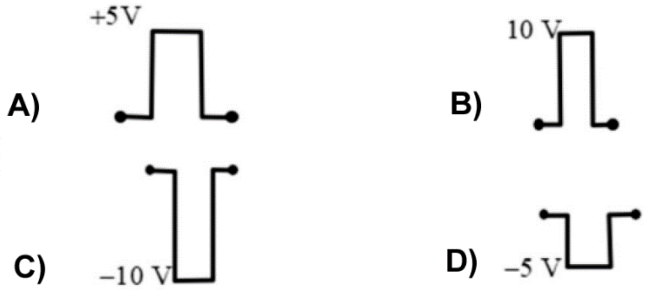


A) 37.5N B) 27.5N C) 17.5N D) 7.5N

11. If in a $p-n$ junction diode, a square input signal of 10 V applied as shown



Then the output signal across R_L will be



12. Time for 20 oscillations of a pendulum is measured as t_1 = 39.6 s; t_2 = 39.9s and t_3 = 39.5 s. What is the least count and mean absolute error of the measurement?

(A) 0.1 s and 0.2 s respectively (B) 0.2 s and 0.1 s respectively

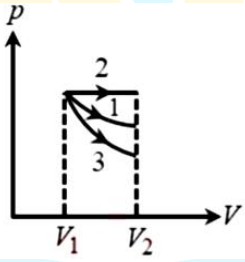
(C) 0.1 s and 0.4 s respectively (D) 0.2 s and 0.3 s respectively

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13. A particle of mass 2m initially at rest and placed on smooth ground explodes into two equal fragments. After explosion, they move under gravity. If after explosion difference between maximum heights of them (measured from ground) be H and difference between time taken to reach maximum height be T then impulse imparted from ground during explosion is –

(A) $\frac{mH}{T}$ (B) $\frac{mH}{2T}$ (C) $\frac{2mH}{T}$ (D) $\frac{3mH}{T}$

14. Starting with the same initial conditions, an ideal gas expands from volume V_1 *to* V_2 in three different ways. The work done by the gas is W_1 if the process is isothermal, W_2 if isobaric and W_3 if adiabatic, then



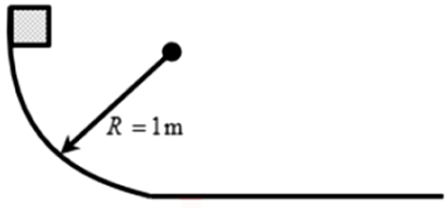
A) $W_2 > W_1 > W_3$ B) $W_2 > W_3 > W_1$

C) $W_1 > W_2 > W_3$ D) $W_1 > W_3 > W_2$

15. n-moles of He gas undergoes a process PV^2 = constant. Initially the pressure and the volumes were P_1 and V_1 and at the end of process rms speed of gas molecule is double compared to its initial rms speed. Then find heat supplied to the gas during the process.

(A) 3 P_1V_1 (B) 2.5 P_1V_1 (C) 1.5 P_1V_1 (D) 2 P_1V_1

16. A book of mass 1 kg slides down a curved track that is one quadrant of a circle of radius 1 m. Its speed at the button is $2ms^{-1}$. The work done by the frictional force is



A) 8 J B) – 8J C) 4 J D) – 4J

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17. (1) Width of depletion layer decreases on increasing doping concentration

(2) Width of depletion layer decreases in forward bias

(3) Width of depletion layer decreases in reverse bias

(4) Width of depletion layer increases with decrease in temperature

(5) Width of depletion layer is more in Zener diode in comparison to normal diode.

(A) 1, 3 and 5 (B) 1, 3, 4 and 5 (C) 1, 2 and 4 (D) 1, 2, 4 and 5

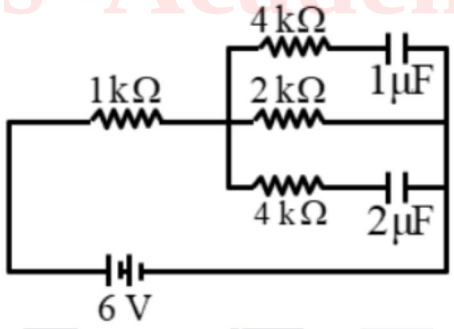
18. The potential energy of 1 kg particle free to move along the $x-axis$ is given by

$$U(x)=\left(\frac{x^4}{4}-\frac{x^2}{2}\right)J$$

The total mechanical energy of the particle is 2 J then, the maximum speed in $\left(ms^{-1}\right)$ is

A) 2 B) $\frac{3}{\sqrt{2}}$ C) $\sqrt{2}$ D) $\frac{1}{\sqrt{2}}$

19. What are the charges stored in the $1\mu F$ and $2\mu F$ capacitors in the circuit below, once the current becomes steady



A) $8\mu C$ *and* $4\mu C$ B) $4\mu C$ *and* $8\mu C$

C) $3\mu C$ *and* $6\mu C$ D) $6\mu C$ *and* $3\mu C$

20. In a concave mirror, an object is placed at a distance x_1 from focus, and image is formed at a distance x_2 from focus. Then focal length of mirror is

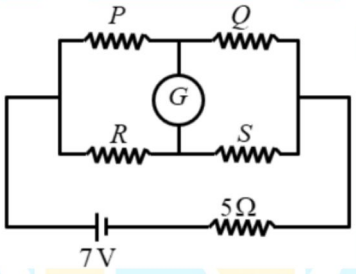
A) $\sqrt{x_1x_2}$ B) $\frac{x_1-x_2}{2}$ C) $\frac{x_1+x_2}{2}$ D) $\sqrt{\frac{x_1}{x_2}}$

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. The resistance of the four arms P, Q, R and S in a Wheatstone's bridge are 10 ohm, 30 ohm, 30 ohm and 90 ohm, respectively. The e.m.f and internal resistance of the cell are 7 volt and 5 ohm respectively. If the galvanometer resistance is 50 ohm, the current drawn from the cell is I, then value of 10 I is___



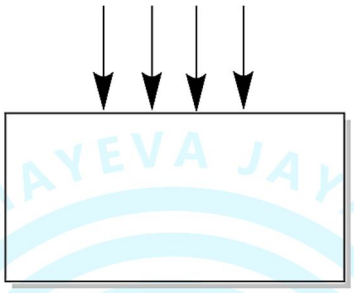
22. A plane wave of monochromatic light falls normally on a uniform thin film of oil which covers a glass plate. The wavelength of source can be varied continuously. Complete constructive interference of reflected lights from air-thin film interface and thin film – glass interface is observed for λ_1 = 5000 Å and λ_2 = 10000 Å and for no other wavelength in between. If refractive index of oil is 1.25 and that of glass is 1.5, the thickness of film (in nm) is

23. A brass rod is 4.000 cm in diameter at 30°C. A steel ring has an inner diameter of 3.998 cm at 30°C. In order that the ring just slides onto the brass rod, the common temperature of the two in absolute value will be _____ °C. [Give your answer in nearest integer] (α_{steel}= 10 × 10^{-6}/°C and β_{brass} = 20 × 10^{-6}/°C)

24. In hydrogen atom, the binding energy of an electron in the ground state is E_1, then the frequency of revolution of the electron in the nth orbit is $\frac{xE_1}{n^y h}$. Here x and y are integers. Find the value of x^y.

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25. A parallel beam of light incidents normally on a glass slab. If 20% light reflects, 30% is transmitted and remaining is absorbed by slab then force due to beam on slab is $x \times 10^{-9}$ N then x = ____ (power of incident beam is 200W).

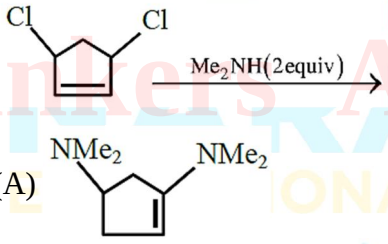


CHEMISTRY MAX.MARKS: 100

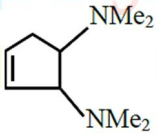
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

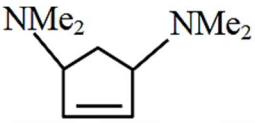
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
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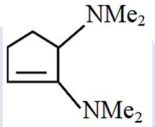
26. Find out the major product formed from the following reaction. [Me: $-CH_3$]



(A)

(B) 

(C) 

(D) 

27. **Statement (I):** Dimethyl glyoxime forms a six membered covalent chelate when treated with $NiCl_2$ solution in presence of NH_4OH.

Statement (II) : Prussian blue precipitate contains iron both in (+2) and (+3) oxidation states.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

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(C) Both Statement I and Statement II are false

(D) Statement I is true but Statement II is false

28. Solubility of aluminium hydroxide (molecular mass = M) in water is **W g per 100 mL** at 25 °C. Its solubility product at 25 °C will be approximately:

(A) $10^3\left(\frac{W}{M}\right)^4$ (B) $10^7\left(\frac{W}{M}\right)^3$ (C) $10^5\left(\frac{W}{M}\right)^4$ (D) $10^5\left(\frac{W}{M}\right)^3$

29. Match the Column.

LIST-I (Pair of compounds)		**LIST-II (Can be distinguished by)**	
(P)	CH_3CHO and PhCHO	(I)	Iodoform test
(Q)	CH_3CH_2OH and CH_3OCH_3	(II)	Barfoed test
(R)	Cane sugar and galactose	(III)	Carbylamine reaction
(S)	$CH_3CH_2NH_2$ and $PhNH_2$	(IV)	Victor meyer test
		(V)	Azo–dye test

(A) P–IV, V ; Q–IV, R–III ; S–V (B) P–I, IV ; Q–I, R–II ; S–III

(C) P–I ; Q–I, IV, R–II ; S–V (D) P–IV ; Q–IV, R–I ; S–III

30. **Statement-1:** In chromatographic technique retardation factor (R_f) does not depend on the polarity of solvent.

Statement -2: Thin layer chromatography is an example of absorption chromatography.

(A) Both statements are true

(B) Statement-1 is true but statement -2 is false

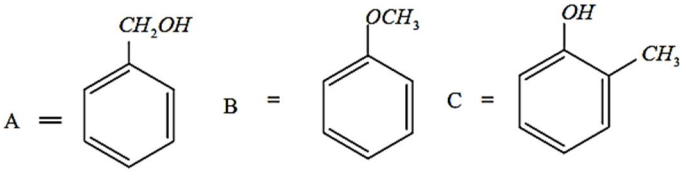
(C) Statement-2 is true but Statement-1 is false

(D) Both Statements are false

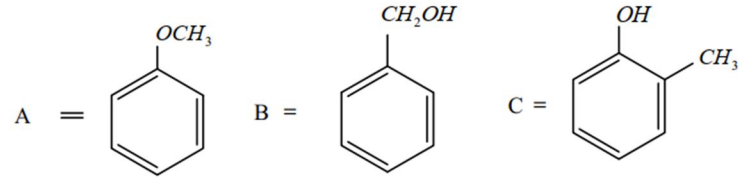
31. Three isomeric compounds A,B,C having molecular formula C_7H_8O show following properties.

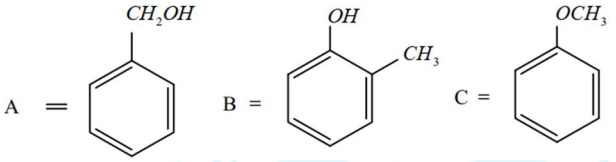
i) Compound B is soluble in aqueous NaOH but A and C are insoluble,

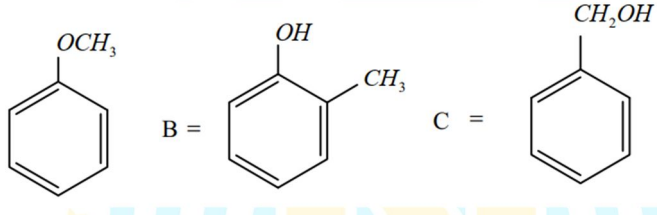
ii) Boiling point order is A > B > C The possible structures of A,B,C are

(A) 

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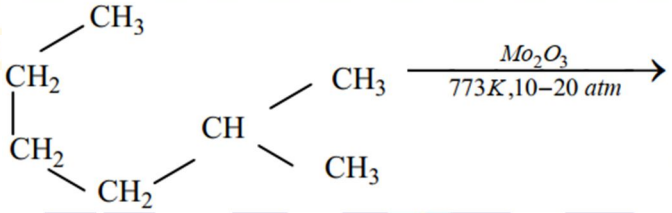
(B) 

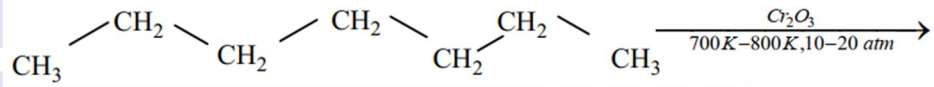
(C) 

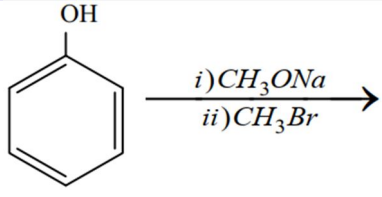
(D) 

32. The major aromatic product of which of the following reactions give only one type of product on mono chlorination with $Cl_2/FeCl_3$

(A) $CH_3 - CH = CHBr \xrightarrow[ii)\,Red\ hot\ iron\ tube, 873K]{i)\,NaNH_2}$

(B) 

(C) 

(D) 

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33. For the reaction: $2H_2 + 2NO \rightleftarrows N_2 + 2H_2O$

the following mechanism has been suggested :

$2NO \rightleftarrows N_2O_2 \left[\text{equilibrium Constant } = K_1(\text{fast})\right]$

$N_2O_2 + H_2 \xrightarrow{K_2} N_2O + H_2O(\text{Slow})$

$N_2O + H_2 \xrightarrow{K_3} N_2 + H_2O(\text{fast})$

K_2 and K_3 are rate constants

According to following mechanism is correctly matched:

(A) $-\frac{1}{2}\frac{d[H_2]}{dt} = K[NO][H_2]$ Where $K=K_2 \times K_3$

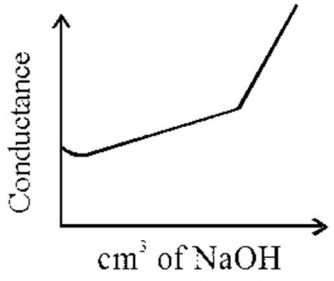
(B) $-\frac{1}{2}\frac{d[NO]}{dt} = K[NO][H_2]$ Where $K=K_2 \times K_1$

(C) $\frac{d[N_2]}{dt} = K[NO][H_2]$ Where $K=K_2 \times K_1$

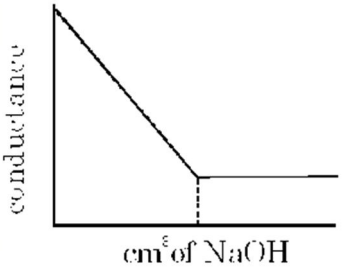
(D) $-\frac{1}{2}\frac{d[NO]}{dt} = K[NO]^2[H_2]$ Where $K=K_2 \times K_1$

34. Which of the following curve correctly represents the titration of CH_3COOH with NaOH solution

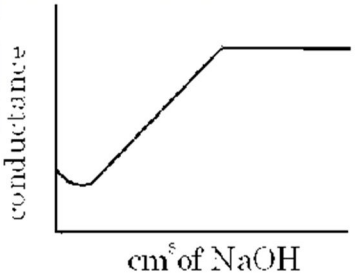
(A)



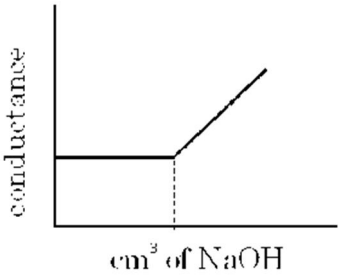
(B)



(C)



(D)



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35. Which reagent can not be used to distinguish t-Butyl alcohol and isopropyl alcohol:

(A) $KMnO_4$ (B) HCl + Anhy $ZnCl_2$

(C) Ammonical $AgNO_3$ (D) $K_2Cr_2O_7$

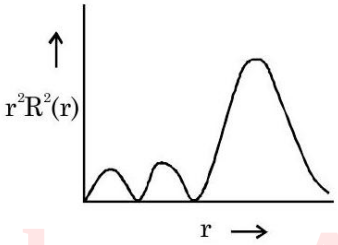
36. The number of unpaired electrons present in the following complexes respectively that explain the magnetic behaviour of those complexes are___

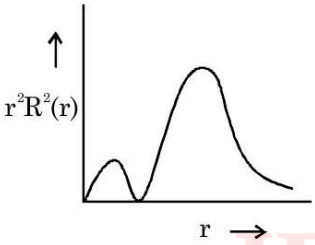
$[Fe(CN)_6]^{4-}, [Fe(H_2O)_6]^{2+}, [Co(NH_3)_6]^{3+}, [MnF_6]^{3-}$

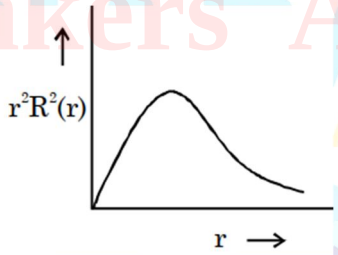
(A) 0, 2, 0, 4 (B) 2, 4, 2, 2

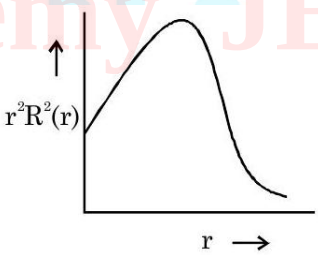
(C) 0, 4, 0, 4 (D) 2, 4, 0, 2

37. Which of the following radial distribution graph corresponds to $l = 2$ for H-atom having least value of 'n' for which $l = 2$ is allowed

(A) 

(B) 

(C) 

(D) 

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38. Incorrectly matched among the following is

	LIST-I (Reaction)	LIST-II (Major product)
(1)	$2CH_3-\overset{\overset{O}{\|\|}}{C}-CH_3 \xrightleftharpoons[\text{(in soxhlet apparatus)}]{Ba(OH)_2}$	$(CH_3)_2C(OH)CH_2COCH_3$ Diacetone alcohol
(2)	$2CH_3COCH_3 \xrightleftharpoons{HCl}$	$(CH_3)_2C = CHCOCH_3$ Mesityl oxide
(3)	$2CH_3COCH_3 \xrightleftharpoons[(ii)\ CH_3COCH_3]{(i)\ HCl}$	$(CH_3)_2C = CHCOCH=C(CH_3)_2$ Phorone
(4)	$CH_3CHO + CH_3COCH_3 \xrightleftharpoons{NaOH}$	$CH_3CH(OH)CH_2COCH_3$ 4-Hydroxypentan-2-one

(A) 1 (B) 2 (C) 3 (D) 4

39. Which of the following order is NOT correct, against mentioned properties?

(A) Sc > Y > La (Ionisation energy) (B) $Hf^{2+} < Hf \simeq Zr$(size)

(C) Aℓ > Si (Electronegativity) (D) Ga > Aℓ (Ionization energy)

40. 100 ml, 1M H_2SO_4 solution of density 1.5 gm/ml is mixed with 400 ml of water, resulting in a diluted solution of density 1.25 gm/ml. What is the molarity of final diluted solution. (Density of water = 1 gm/ml).

(A) 0.2 M (B) 0.1 M (C) 0.3 M (D) 0.4 M

41. For the Homogeneous 1st order gaseous reaction

$A \longrightarrow B + C + D + E + ... + X + Y + Z.$

The initial pressure was P_0 while after time t, total pressure was P. The expression for rate constant k in terms of P_0, P and t is :

(A) $k = \frac{2.303}{t}\log\frac{P_0}{P_0 - P}$ (B) $k = \frac{2.303}{t}\log\frac{25P_0}{26P_0 - P}$

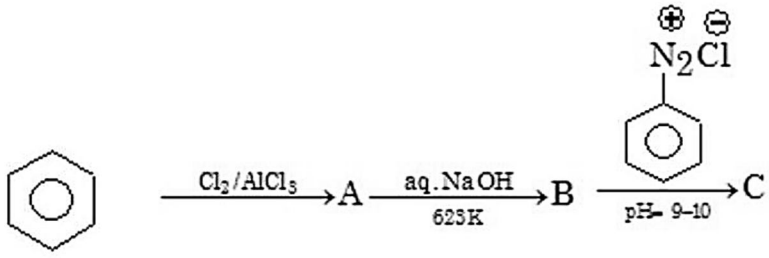
(C) $k = \frac{2.303}{t}\log\frac{24P_0}{25P_0 - P}$ (D) $k = \frac{2.303}{t}\log\frac{24P_0}{P_0 - P}$

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42. The orbital angular momentum of electrons in a state of n = 4 and m = – 3 is

(A) 0 (B) $\frac{h}{\sqrt{2}\pi}$ (C) $\frac{\sqrt{6}h}{2\pi}$ (D) $\frac{\sqrt{3}h}{\pi}$

43. What is incorrect statement regarding reaction sequence :



(A) B gives violet colour complex with neutral $FeCl_3$

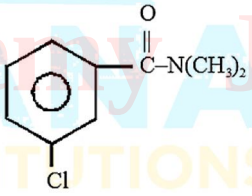
(B) C is orange colour dye.

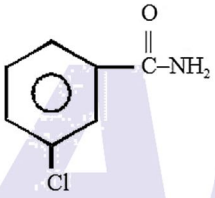
(C) Benzene is less reactive than product A for electrophillic aromatic substitution reaction.

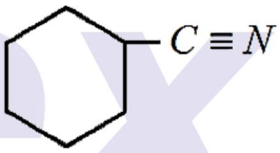
(D) Formation of C from B is coupling reaction.

44. Out of the following which can be converted into amine in presence of NaOBr.

(A) 

(B) 

(C) 

(D) 

45. The correct solubility order is

(A) $As_2S_3 > NiS$ (B) $CoS > CuS$

(C) $ZnS < PbS$ (D) $SnS > (NH_4)_2S$

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. From the following data at 25°C, calculate the magnitude of enthalpy of reaction (in kJ/mol) (nearest integer)

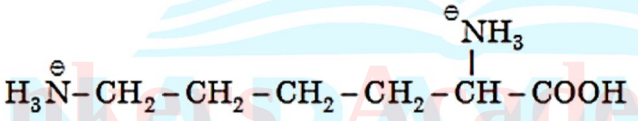
$$Fe_2O_{5(s)} + 3C_{(graphite)} \longrightarrow 2FeO_{(s)} + 3CO_{(g)}$$

Given data of enthalpy of formation is :

$\Delta_f H°Fe_2O_5(s) = -824.5\,kJ\,mol^{-1}$,
$\Delta_f H°FeO(s) = -65\,kJ\,mol^{-1}$,
$\Delta_f H°CO(g) = -110.5kJ\,mol^{-1}$

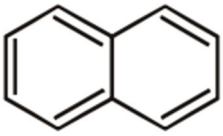
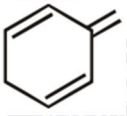
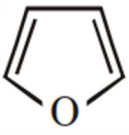
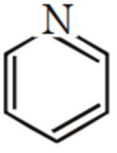
$\Delta H_{atom}C_{(graphite)} = 716.67kJ\ mol^{-1}$

47. The structure of basic amino acid is given as follows :



The $Pk_{a_1}, Pk_{a_2}, Pk_{a_3}$ value are respectively 2.18, 8.95 and 10.53. If the isoelectric point of the given amino acid is x. Find 100x.

48. Number of aromatic compounds in the following

(1)		(2)		(3)		(4)	
(5)		(6)		(7)		(8)	

49. A polyhydroxy compound having molecular mass 150, on reacting the compound with CH_3COCl, the precipitate obtained has molecular mass of 318. Calculate number of OH group present in compound :

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50. In sulphur estimation 0.470 g of an organic compound gave 2.33 g of barium sulphate. The percentage of sulphur in the compound is ……..(nearest integer)

MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The locus of the vertices of the family of parabolas $y=\frac{a^3x^2}{3}+\frac{a^2x}{2}-2a$ is $(a\in R)$

(A) $xy=\frac{105}{64}$ (B) $xy=\frac{3}{4}$ (C) $xy=\frac{35}{16}$ (D) $xy=\frac{64}{105}$

52. The value of $\tan\left(\sum_{r=1}^{\infty}\cot^{-1}\left(\frac{r^2}{2}+\frac{15}{8}\right)\right)$ is equal to

(A) 1 (B) 2 (C) 3 (D) 4

53. Let $x^2+y^2+Ax+By+C=0$ be a circle passing through (0, 6) and touching the parabola $y=x^2$ at (2, 4). Then, A + B + C is equal to _____.

(A) 16 (B) $\frac{-36}{5}$ (C) $\frac{36}{5}$ (D) –8

54. Let $f:R\to R$ be a function defined by $f(x)=x^3+x-1$ and 'g' is the inverse of f, then the value of $\frac{g(9)}{g'(9)}$ is (g' is the derivative of g)

(A) 26 (B) 32 (C) 11 (D) 17

55. The value of $\left(4\cos^2 9^0-1\right)\left(4\cos^2 27^0-1\right)\left(4\cos^2 81^0-1\right)\left(4\cos^2 243^0-1\right)$ is

(A) –1 (B) 1 (C) 2 (D) –2

56. If $P=\begin{bmatrix}\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}}\end{bmatrix}$, $A=\begin{bmatrix}1 & 1\\ 0 & 1\end{bmatrix}$ and $Q=PAP^T$, then $P^T\left(Q^{2005}\right)P$ is equal to

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(A) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} \frac{\sqrt{3}}{2} & 2005 \\ 1 & 0 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 2005 \\ \frac{\sqrt{3}}{2} & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & \frac{\sqrt{3}}{2} \\ 0 & 2005 \end{bmatrix}$

57. Let from a point A (h, k), chords of contact are drawn to the ellipse $x^2 + 2y^2 = 6$ where all these chord touch the ellipse $x^2 + 4y^2 = 4$. Then, the perimeter (in units) of the locus of point A is

(A) 2π (B) 3π (C) 4π (D) 6π

58. The value of x for which 3rd term in the expansion of $\left(\sqrt{x} + \frac{\sin x}{x}\right)^6$ is $\frac{15}{4}$ is $\left(0 < x < \frac{\pi}{2}\right)$.

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{12}$

59. Let f(x) is continuous for $x \in R$ except at x = 0 such that $f'(x) < 0 \ \forall \ x \in (-\infty, 0)$ and $f'(x) > 0 \ \forall \ x \in (0, \infty)$. Also given that $\lim\limits_{x \to 0^-} f(x) = 4$, $\lim\limits_{x \to 0^+} f(x) = 3$ and $f(0) = 4$ then $\lim\limits_{x \to 0} f\left(\cos^3 x - \cos^2 x\right) + \lim\limits_{x \to 0} f\left(\sin^2 x - \sin^3 x\right) =$

(A) 4 (B) 3 (C) 6 (D) 7

60. Let $g : [1, \infty) \to [2, \infty)$ be a differentiable function satisfying $6\int_1^x g(t)dt = 3xg(x) - x^3 - 5 \forall x \geq 1$ then g(2) = _____

(A) 4 (B) 5 (C) 1 (D) 6

61. Let the straight line x = b divide the area enclosed by $y = (1 - x)^2$, y = 0, and x = 0 into two parts R_1 ($0 \leq x \leq b$) and R_2 ($b \leq x \leq 1$) such that $R_1 - R_2 = \frac{1}{4}$. Then b equals

A) $\frac{3}{4}$ B) $\frac{1}{2}$ C) $\frac{1}{3}$ D) $\frac{1}{4}$

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62. The surface area of a balloon of spherical shape being inflated, increases at a constant rate. If initially, the radius of balloon is 3 units and after 5 seconds, it becomes 7 units, then its radius after 9 seconds is :

(A) 9 (B) 10 (C) 11 (D) 12

63. If $\{a_i\}_{i=1}^{n}$ where n is an even integer is an arithmetic progression with common difference 1, and $\sum_{i=1}^{n} a_i = 192$, $\sum_{i=1}^{n/2} a_{2i} = 120$ then n is equal to

(A) 48 (B) 96 (C) 92 (D) 104

64. If $f(x) = 4\sin^2 x + 6\sin x\cos x + 12\cos^2 x$ & $g(x) = \left[\frac{f(x)}{k \cdot sqn\left(x^2+1\right)}\right]$,[·] is G.I.F function, then min integral value of k So that g(x), is continuous is

(A) 12 (B) 11 (C) 14 (D) 13

65. If A is a square matrix of order 3 such that $\det(A) = 3$ and $\det\left(adj\left(-4adj\left(-3adj\left(3adj\left((2A)^{-1}\right)\right)\right)\right)\right) = 2^m 3^n$. Then m + 2n is equal to:

(A) 2 (B) 3 (C) 6 (D) 4

66. Let L_1 and L_2 be the lines whose equation are $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ respectively. A and B are two points on L_1 and L_2 respectively such that AB is perpendicular to both the lines L_1 and L_2. Find distance between the points A and B

A) $\sqrt{30}$ B) $2\sqrt{30}$ C) $3\sqrt{30}$ D) $4\sqrt{30}$

67. The value of the integral $\int_{1/2}^{2} \frac{\tan^{-1} x}{x} dx$ is equal to

(A) $\frac{1}{2}\log_e 2$ (B) $\pi\log_e 2$ (C) $\frac{\pi}{4}\log_e 2$ (D) $\frac{\pi}{2}\log_e 2$

68. Let $A = \{x \in R : [x+3]+[x+4] \le 1\}$, $B = \left\{x \in R : 3^x \left(\sum_{r=1}^{\infty} \frac{3}{10^r}\right)^{x-3} < 3^{-3x}\right\}$, where [t] denotes greatest integer function. Then,

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(A) $A \subset B$ (B) $A \cap B = \phi$ (C) $A = B$ (D) $B \subset A$

69. If y(x) satisfies the differential equation $y' - y \tan x = 2x \sec x$ and $y(0) = 0$, then

A) $y\left(\frac{\pi}{3}\right) = \pi^2$ B) $y'\left(\frac{\pi}{4}\right) = \frac{\pi^2}{18}$

C) $y\left(\frac{\pi}{3}\right) = \frac{\pi^2}{9}$ D) $y'\left(\frac{\pi}{3}\right) = \frac{4\pi}{3} + \frac{2\pi^2}{3\sqrt{3}}$

70. The set of possible values of λ for which $x^2 - (\lambda^2 - 5\lambda + 5)x + (2\lambda^2 - 3\lambda - 4) = 0$ has roots, whose sum and product are both less than 1, is

A) $\left(-1, \frac{5}{2}\right)$ B) (1, 4) C) $\left[1, \frac{5}{2}\right]$ D) $\left(1, \frac{5}{2}\right)$

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
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71. $\lim_{x \to 0} \frac{\int_0^x \left(t^2 + e^{t^2}\right)^{\frac{1}{1-\cos t}} dt}{\left(e^x - 1\right)} = e^k$. Then k is

72. Let A and B be independent events such that P(A) = p, P(B) = 2p. The largest value of p, for which P (exactly one of A, B occurs) $= \frac{5}{9}$, is λ, then $12\lambda =$

73. How many 3×3 matrices M with entries from {0, 1, 2} are there, for which the sum of the diagonal entries of $M^T M$ is 5?

74. The number of values of k for which the linear equations posses a non-zero solution is: $4x+ky+2z=0$, $kx+4y+z=0$, $2x+2y+z=0$

75. If all the words (with or without meaning) having five letters, formed using all the letters of the word SMALL and arranged as in a dictionary; then the position of the word SMALL is

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